

Rapidity frames in p Pb and PbPb ultraperipheral collisions: the $\Delta y = 0.465$ shift

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1 Summary

In p Pb collisions at the LHC the two beams carry different energies per nucleon, so the nucleon–nucleon (NN) centre-of-mass frame moves in the laboratory with rapidity $|\Delta y| = \frac{1}{2} \ln(208/82) \simeq 0.465$ towards the proton. Theory calculations (this code, and Guzey, Innocenti, Stařto and Strikman, arXiv:2606.05469) are done in the NN centre-of-mass frame, while detectors measure laboratory rapidities. Comparing the two therefore requires shifting by 0.465. In PbPb both beams have the same energy per nucleon, the centre-of-mass frame is at rest in the laboratory, and no shift is needed.

2 Origin of the shift

The LHC bends both beams with the same magnets on the same orbit, so both have the same momentum per unit charge (magnetic rigidity). Per nucleon, this gives

$$E_{\text{Pb}}/\text{nucleon} = \frac{Z}{A} E_p. \quad (1)$$

For the 8.16 TeV p Pb run: $E_p = 6.5$ TeV and $E_{\text{Pb}}/\text{nucleon} = 6.5 \times 82/208 \simeq 2.56$ TeV. For ultrarelativistic beams,

$$\sqrt{s_{NN}} = 2\sqrt{E_p E_{\text{Pb}}} \simeq 2\sqrt{6.5 \times 2.56} \text{ TeV} \simeq 8.16 \text{ TeV}, \quad (2)$$

the value used in the code (`param.ss = 8160` for `TARGET=pA`).

Take the lead beam along $+z$ and the proton along $-z$. The total light-cone momenta per nucleon pair are $P^+ \propto E_{\text{Pb}}$ and $P^- \propto E_p$, so the rapidity of the NN centre of mass in the laboratory is

$$Y_{\text{CM}} = \frac{1}{2} \ln \frac{P^+}{P^-} = \frac{1}{2} \ln \frac{E_{\text{Pb}}}{E_p} = -\frac{1}{2} \ln \frac{A}{Z} = -\frac{1}{2} \ln \frac{208}{82} \simeq -0.465. \quad (3)$$

The centre of mass moves towards the proton, the beam with more momentum per nucleon.

Rapidity is additive under boosts along the beam axis, so every particle’s rapidity changes by the same constant between the two frames:

$$y_{\text{lab}} = y_{\text{CM}} + Y_{\text{CM}}. \quad (4)$$

The shape of $d\sigma/dy$ is unchanged; the whole distribution is only translated by 0.465 units.

3 Why the calculation lives in the centre-of-mass frame

The kinematics in the code assume a symmetric NN frame:

- The photon energy fraction $z = 2\omega/\sqrt{s_{NN}}$ uses the lead Lorentz factor $\gamma = \sqrt{s_{NN}}/(2m_N)$. This is the lead’s boost in the NN centre-of-mass frame, not in the laboratory.
- The D^0 rapidity enters through $p^+ = m_T e^y/\sqrt{2}$, and the smallest reachable photon energy fraction is $z_{\min} = m_T e^y/\sqrt{s_{NN}}$ (`src/main.cpp`). Both are centre-of-mass quantities.
- The momentum fractions probed in the target, $x \sim (m_T/\sqrt{s_{NN}}) e^{-y}$, are likewise centre-of-mass expressions.

So the y passed to D0 and written to the output files is y_{CM} .

4 Why PbPb needs no shift

In PbPb both beams have the same Z/A , hence the same energy per nucleon (2.68 TeV each at $\sqrt{s_{NN}} = 5.36$ TeV). Then $P^+ = P^-$, $Y_{\text{CM}} = 0$ and $y_{\text{lab}} = y_{\text{CM}}$. The same holds for pp . The shift appears only for asymmetric systems ($p\text{Pb}$, $p\text{O}$, O-Pb , $\dots$), where $Z_1/A_1 \neq Z_2/A_2$.

System	$E_1/\text{nucleon}$	$E_2/\text{nucleon}$	$ Y_{\text{CM}} $
PbPb, 5.36 TeV	2.68 TeV	2.68 TeV	0
$p\text{Pb}$, 8.16 TeV	6.5 TeV (p)	2.56 TeV (Pb)	$\frac{1}{2} \ln(208/82) \simeq 0.465$

5 Sign conventions

The size of the shift is fixed; its sign depends on which beam defines $+y$. There are two conventions to keep apart.

This code and arXiv:2606.05469. Positive rapidity is the direction of the photon-emitting lead nucleus; the proton target moves towards negative rapidity. In the code this follows from $z_{\min} \propto e^y$: larger y requires a more energetic photon, i.e. the D^0 goes along the photon (lead) direction. The paper states the same convention explicitly in Sec. 6 and in the captions of its $p\text{Pb}$ figures (“the nucleus moves in the positive-rapidity direction and the proton in the negative-rapidity direction”).

Experiments. LHC experiments usually take the proton direction as $+y$ and often quote the two beam configurations ($p\text{-Pb}$ and $\text{Pb-}p$) separately. Each measurement must be checked for its own convention.

Conversion. With the proton along $+y$ in the experiment, a measured laboratory rapidity $y_{\text{lab}}^{\text{exp}}$ corresponds to

$$y_{\text{CM}}^{\text{exp}} = y_{\text{lab}}^{\text{exp}} - 0.465 \quad (\text{proton still along } +y), \quad (5)$$

and, flipping to the code’s orientation (lead along $+y$),

$$y_{\text{code}} = -(y_{\text{lab}}^{\text{exp}} - 0.465). \quad (6)$$

A data bin $a < y_{\text{lab}}^{\text{exp}} < b$ therefore corresponds to $-(b - 0.465) < y_{\text{code}} < -(a - 0.465)$.

6 Does arXiv:2606.05469 apply the shift?

No. In the LaTeX source of Guzey, Innocenti, Staśto and Strikman, “Diffractive open charm photo-production in ultraperipheral lead-lead and proton-lead collisions at the LHC” (arXiv:2606.05469), the $p\text{Pb}$ section gives predictions at $\sqrt{s_{NN}} = 8.16$ TeV for $-2 < y < 2$ with the nucleus along $+y$. The rapidity shift, laboratory versus centre-of-mass frames and the value 0.465 are never mentioned. Their y is therefore the NN centre-of-mass rapidity, the same as ours.

This is consistent: a theory prediction does not need the shift on its own. Consequences for this project:

- Our $p\text{Pb}$ output (`run_pA.sh`) can be compared with their $p\text{Pb}$ figures directly, bin by bin, with no shift, since both use the centre-of-mass frame with lead along $+y$.
- The shift is needed only when comparing either calculation with measured $p\text{Pb}$ data quoted in laboratory rapidity.

7 Which nucleus emits the photon

The photon flux scales as Z^2 , so in $p\text{Pb}$ UPCs the lead emits essentially all photons and the proton acts as the target; the proton-emitted contribution ($Z = 1$) is suppressed by $1/82^2$ and neglected. The distribution is then asymmetric in y , which is why the orientation above matters. In PbPb either nucleus can emit, the measured distribution is symmetric in y , and the physical cross section is the sum of the two emitter configurations, evaluated at $+y$ and $-y$.